Atlas 145: Concentrating Two Triangle Pages on an Edge of a 4-Cycle

Abstract

We prove the chromatic-polynomial lower bound for NetworkX Atlas graph 145 on the full interval $p \in [1/2, 1]$ and for every graphon. The graph is a 4-cycle with two triangle pages on one edge. Moving one page to the adjacent cycle edge produces Atlas 148, whose full-interval theorem is known. The difference of the two homomorphism densities is exactly one half of a square under the 4-cycle homomorphism measure. This gives a short, division-free transfer valid on arbitrary probability spaces.

1 The two page placements

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and jointly measurable. Put

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu^2, \qquad S(x,y) = \int_{\Omega} W(x,z)W(y,z)\,\mathrm{d}\mu(z).$$

On labelled vertices $0,1,2,3,4,5$, let H_{same} have edges

$$01, 12, 23, 30, 04, 14, 05, 15. \tag{1}$$

Thus 0–1–2–3–0 is a 4-cycle and vertices $4, 5$ are two triangle pages on the same cycle edge 01. This graph is NetworkX Atlas 145, with graph6 code `Exe_`. Let H_{adj} instead have edges

$$01, 12, 23, 30, 04, 14, 05, 35. \tag{2}$$

It has one triangle page on each of the adjacent cycle edges 01 and 03. This is Atlas 148, with graph6 code `EyUG`. The exact isomorphism checks are reproduced by the verifier cited below.

The triangle in H_{same} forces chromatic number at least three. A proper three-colouring has colour classes

$$\{0,2\}, \qquad \{1,3\}, \qquad \{4,5\}.$$

Hence the chromatic number is three and the admissible density interval is $p \in [1/2, 1]$.

To count its proper r-colourings, first colour the page-bearing edge 01. Each page then has $r-2$ choices. Conditional on the two distinct endpoint colours, the length-three path 0–3–2–1 has

$$(r-1) + (r-2)^2 = r^2 - 3r + 3$$

proper internal colourings. Consequently

$$\chi_{H_{\mathrm{same}}}(r) = r(r-1)(r-2)^2(r^2-3r+3). \tag{3}$$

The same polynomial holds for H_{adj}, either by the analogous direct count or by the computation in the Atlas 148 theorem. Substitution into the chromatic target gives

$$\Phi_{H_{\mathrm{same}}}(p) = (1-p)^6 \chi_{H_{\mathrm{same}}}\left(\frac{1}{1-p}\right) = p(2p-1)^2(3p^2-3p+1). \tag{4}$$

2 The page-concentration square

For $x_0, x_1, x_2, x_3 \in \Omega$, define the finite nonnegative measure

$$\mathrm{d}\Lambda = W(x_0, x_1)W(x_1, x_2)W(x_2, x_3)W(x_3, x_0)\,\mathrm{d}\mu(x_0)\,\mathrm{d}\mu(x_1)\,\mathrm{d}\mu(x_2)\,\mathrm{d}\mu(x_3). \tag{5}$$

No normalization is imposed; its total mass may be zero. Put

$$A = S(x_0, x_1), \qquad B = S(x_0, x_3).$$

Lemma 2.1 (Adjacent-page concentration)**.** *For every graphon on every probability space,*

$$t(H_{\mathrm{same}}, W) - t(H_{\mathrm{adj}}, W) = \frac{1}{2}\int_{\Omega^4} (A - B)^2\,\mathrm{d}\Lambda \geq 0. \tag{6}$$

Proof. Integrating the two page vertices in (1) and (2) gives, respectively,

$$t(H_{\mathrm{same}}, W) = \int A^2\,\mathrm{d}\Lambda, \qquad t(H_{\mathrm{adj}}, W) = \int AB\,\mathrm{d}\Lambda.$$

The reflection of the 4-cycle that interchanges x_1 and x_3 while fixing x_0, x_2 preserves (5). Therefore

$$\int A^2\,\mathrm{d}\Lambda = \int B^2\,\mathrm{d}\Lambda.$$

Expanding the square now gives

$$\frac{1}{2}\int (A - B)^2\,\mathrm{d}\Lambda = \int A^2\,\mathrm{d}\Lambda - \int AB\,\mathrm{d}\Lambda,$$

which is (6). Every integrand is bounded and measurable, so Tonelli–Fubini and the measure-preserving variable permutation are valid on an arbitrary probability space. No quotient or positivity assumption on $t(C_4, W)$ is used. □

The identity also records its precise equality condition: the two rooted common-neighbour values agree Λ-almost everywhere. We do not need to classify the graphons satisfying that condition.

3 The Atlas 145 bound

We use the following already established result.

Theorem 3.1 (Atlas 148 theorem [1])**.** *Every graphon of edge density $p \in [1/2, 1]$ satisfies*

$$t(H_{\mathrm{adj}}, W) \geq p(2p - 1)^2(3p^2 - 3p + 1).$$

That theorem is an arbitrary-graphon result. Its low-density half uses the K_3 case of Reiher's clique density theorem [2]; its high-density half uses a two-coordinate Hilbert projection and exact rational Bernstein certificates. No finite-step or regularity hypothesis is imported here.

Corollary 3.2 (Atlas 145, graph6 `Exe_`)**.** *Every graphon W of edge density $p \in [1/2, 1]$ satisfies*

$$\boxed{t(H_{\mathrm{same}}, W) \geq p(2p - 1)^2(3p^2 - 3p + 1) = \Phi_{H_{\mathrm{same}}}(p).}$$

Proof. Lemma 2.1 and Theorem 3.1 give

$$t(H_{\mathrm{same}}, W) \geq t(H_{\mathrm{adj}}, W) \geq \Phi_{H_{\mathrm{same}}}(p),$$

where the last target equality follows from (4). At $p = 1/2$ the target is zero, and at $p = 1$ the graphon is one almost everywhere and both sides are one, so the endpoint cases require no limiting argument. □

4 Audit and limitations

The comparison is an exact relocation theorem for these two page placements: two pages on one edge of a 4-cycle dominate one page on each of two adjacent edges. It does not assert arbitrary triangle-page concentration on an arbitrary base graph; the reflection symmetry of the 4-cycle is the exact hypothesis used. In particular, putting the two pages on opposite cycle edges gives Atlas 152 (graph6 `EhNG`), which has an accepted exact counterexample; the present one-sided concentration inequality cannot prove a lower bound for that placement. The argument also does not revive the false product comparison

$$p\,t(H_{\mathrm{same}}, W) \geq t(K_4 - e, W)t(C_4, W)$$

recorded in the failed-attempt ledger.

The Atlas identifications, graph6 codes, chromatic polynomials, targets, density decompositions, and exact square identity on 503 rational step graphons were independently reconstructed in exact rational arithmetic.

No Lean file is changed. A future formalization needs the finite-measure square identity above and the existing Atlas 148 theorem as its only graph-specific inputs. The latter still carries the formal dependency on the sharp triangle-density theorem.

References

[1] Atlas 148 proof note, `notes/atlas148_paw_bias_hilbert_projection.pdf`.

[2] C. Reiher, *The clique density theorem*, Annals of Mathematics **184** (2016), 683–707; `https://annals.math.princeton.edu/2016/184-3/p01`.

A What the Lean formalization actually proves

The corollary is formalized in full, on $p \in [1/2, 1]$. The development is `lean/Taeyoung/Methods/Atlas145.lean`, the catalogue theorem is `satisfiesLowerBound_145`, and every declaration depends on exactly `[propext, Classical.choice, Quot.sound]`. Theorem 3.1 is no longer quoted from outside: it is `Methods.Atlas148.homDensity_graph148_bound`, itself formalized, so the closing remark of Section 4 is superseded on both counts.

A.1 The square identity becomes one Cauchy–Schwarz

Lemma 2.1 is proved above by reflecting the four-fold integral (5) in the map $x_1 \leftrightarrow x_3$. A choice of labelling removes the reflection entirely.

Integrate the two *opposite* corners x_0, x_2 of the 4-cycle outermost. The remaining cycle vertices x_1 and x_3 then appear in disjoint factors of $\mathrm{d}\Lambda$, so for fixed x_0, x_2 everything factorizes through the three arm integrals

$$P = \int_\Omega W(x_0, u)W(u, x_2)\,\mathrm{d}\mu(u), \quad Q = \int_\Omega W(x_0, u)W(u, x_2)S(x_0, u)\,\mathrm{d}\mu(u),$$

$$R = \int_\Omega W(x_0, u)W(u, x_2)S(x_0, u)^2\,\mathrm{d}\mu(u),$$

in the form

$$t(H_{\mathrm{adj}}, W) = \iint Q^2, \qquad t(H_{\mathrm{same}}, W) = \iint P \cdot R.$$

The comparison is then the pointwise weighted Cauchy–Schwarz

$$Q^2 \le P \cdot R,$$

that is $(\int A\eta)^2 \le (\int A)(\int A\eta^2)$ at weight $A(u) = W(x_0,u)W(u,x_2)$ and $\eta(u) = S(x_0,u)$ — the same inequality this project uses throughout, and the only inequality in the whole file. Note that $P = S(x_0,x_2)$ is the codegree of the opposite corners, so all three arms are instances of one operator.

The two formulations agree, of course; (6) is the same statement with the Cauchy–Schwarz deficit written as a square. What the relabelling buys is that no permutation of a four-fold integral has to be justified, and no measure-preserving change of variables is needed: the factorization is ordinary Fubini through the peeling the project already performs for every row.

A.2 Transporting the Atlas 148 bound

Both graphs are laid out on that frame — vertices $0,1$ the opposite corners, $2,3$ the other two cycle vertices, $4,5$ the pages — so the Lean copy of Atlas 148 used here is

$$02,\ 03,\ 04,\ 05,\ 12,\ 13,\ 24,\ 35,$$

which is `Methods.Atlas148.graph148` relabelled. The two densities are identified by `Foundation.homDensity_iso`, so the Atlas 148 theorem applies to the frame copy without being reproved.

A.3 Chromatic data

The polynomial (3) is proved, not assumed. Its factor $r^2 - 3r + 3$ is irreducible over the rationals, so no clique-attachment tower produces it, and the route is the same as for Atlas 148: the six surjective colouring counts

$$0,\ 0,\ 0,\ 18,\ 264,\ 840,\ 720$$

are checked by kernel reduction, the top one through $\mathrm{surj}(n) = n!$, and the target (4) follows by one `field_simp`. The counts coincide with Atlas 148's, as they must, but they are checked again: the two graphs are not isomorphic, so nothing transports.

A.4 Scope

The formalization asserts exactly the corollary: the two stated page placements on a 4-cycle, in that direction. It does not formalize the general relocation principle discussed in Section 4, and — as that section records — the opposite-edge placement is Atlas 152, which has a verified counterexample in this project (`Methods.Negative.Atlas152.violatesLowerBound_152`), so no one-sided concentration inequality could cover it.

With this row the catalogue reaches 93 verified rows: every Atlas graph whose status is currently accepted, positive or negative, is proved in Lean with no `sorry`.